

Build a Mini AI Workflow Builder

Objective

Build a web app where users create simple AI workflows from a system prompt, one or more tools, and some decision logic.

We want to see how you build a configurable AI system, not a single hardcoded agent.

Problem statement

Users should be able to configure a workflow visually and then chat with it.

Each workflow has:

- A system prompt
- One or more tools
- A sequence of steps, or a simple decision flow
- A chat interface to run it

Workflows are saved, so users can edit and reuse them later.

Functional requirements

Workflow builder

Let users:

- Create a new workflow
- Name it
- Set a system prompt
- Enable or disable tools
- Save and edit workflows

Tools

Include at least three tools, for example:

- Web search
- Calculator
- Mock email sender (logs the email instead of sending it)

Add more if you want.

Chat interface

Users should be able to:

- Pick a saved workflow
- Chat with it
- See tool calls and intermediate steps
- Get the final response

Technical requirements

- Any frontend framework.
- Any backend framework.
- Store workflows in a database.
- Any LLM provider.
- It should run locally, with clear setup instructions.

Bonus (optional)

- Drag-and-drop workflow editor
- Streaming responses
- Multiple LLM providers
- Workflow versioning
- Execution history
- Better UI/UX
- Docker setup

Submission

Include:

- The GitHub repository
- A README with setup instructions
- `.env.example`
- A short write-up of your architecture and design decisions

Evaluation criteria

We mainly look at:

- Overall architecture
- AI orchestration
- Code quality
- Frontend implementation
- Backend and persistence
- Extensibility
- User experience
- Completeness over polish

We'd rather see a working end-to-end solution than an incomplete one with advanced features.